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Sam has $800 in his account. He will deposit $1 in his account one week from now, $2 two weeks from now, and each week thereafter he will deposit an amount that is $1 greater than the amount that he deposited one week before. If there are no other transactions, how much money will Sam have in his account 50 weeks from now?

A. $850

B. $1,200

C. $1,675

D. $2,075

E. $3,350

Arithmetic Series and Sequences

Determine the amount Sam will have in his account after 50 weeks if the account starts with $800 and Sam deposits $1 more each week than he deposited the week before.

Sam's deposits over the 50-week period will be 1 +2 + 3 + … + 48 + 49 + 50 = (1 + 50) + (2 + 49) + (3 + 48) + … + (25 + 26) = (25)(51) = $1,275. Thus, after 50 weeks Sam will have $800 + $1,275 = $2,075.

The correct answer is D.

•

A certain state's milk production was 980 million pounds in 2007 and 2.7 billion pounds in 2014. Approximately how many more million gallons of milk did the state produce in 2014 than in 2007? (1 billion = 10^9 and 1 gallon = 8.6 pounds.)

A. 100

B. 200

C. 1,700

D. 8,200

E. 14,800

Arithmetic Measurement Conversion

Using 2.7 billion = 2,700 million, the amount of increase in millions of gallons can be calculated as follows.

The amount of increase is:

(2,700 – 980) million pounds = 1,720 million pounds

To convert pounds into gallons:

$$\left(1,720 \text{ million } \cancel{\text{pounds}}\right)\left(\frac{1 \text{ gallon}}{8.6 \cancel{\text{pounds}}}\right) = 200 \text{ million gallons}$$

The correct answer is B.

•

Working simultaneously and independently at an identical constant rate, 4 machines of a certain type can produce a total of x units of Product P in 6 days. How many of these machines, working simultaneously and independently at this constant rate, can produce a total of $3x$ units of Product P in 4 days?

A. 24

B. 18

C. 16

D. 12

E. 8

Algebra Applied Problems

Define a *machine day* as 1 machine working for 1 day. Then 4 machines each working 6 days is equivalent to (4)(6) =

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Define a *machine day* as 1 machine working for 1 day. Then 4 machines each working 6 days is equivalent to (4)(6) = 24 machine days. Thus, x units of Product P were produced in 24 machine days, and $3x$ units of Product P will require (3)(24) = 72 machine days, which is equivalent to $\frac{72}{4} = 18$ machines working independently and simultaneously for 4 days.

The correct answer is B.

•

The symbol Δ denotes one of the four arithmetic operations: addition, subtraction, multiplication, or division. If 6 Δ 3 ≤ 3, which of the following must be true?

I. 2 Δ 2 = 0

II. 2 Δ 2 = 1

III. 4 Δ 2 = 2

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic Operations with Integers

If Δ represents addition, subtraction, multiplication, or division, then 6 Δ 3 is equal to either 6 + 3 = 9, or 6 – 3 = 3, or 6 × 3 = 18, or 6 ÷ 3 = 2. Since it is given that 6 Δ 3 ≤ 3, Δ represents either subtraction or division.

Statement I is true for subtraction since 2 – 2 = 0 but not true for division since 2 ÷ 2 = 1.

Statement II is not true for subtraction since 2 – 2 = 0 but is true for division since 2 ÷ 2 = 1.

Statement III is true for subtraction since 4 – 2 = 2 and is true for division since 4 ÷ 2 = 2. Therefore, only Statement III must be true.

The correct answer is C.

•

If $mn \neq 0$ and 25 percent of n equals $37\frac{1}{2}$ percent of m, what is the value of $\frac{12n}{m}$?

A. 18

B. $\frac{32}{3}$

C. 8

D. 3

E. $\frac{9}{8}$

Algebra Percents; First-Degree Equations

It is given that $(25\%)n = (37.5\%)m$, or $0.25n = 0.375m$. The value of $\frac{12n}{m}$ can be found by first finding the value of $\frac{n}{m}$ and then multiplying the result by 12. Doing this gives $\frac{12n}{m} = \left(\frac{0.375}{0.25}\right)(12) = 18$. Alternatively, the numbers involved allow for a series of simple equation transformations to be carried out, such as the following:

$$\begin{aligned} 0.25n &= 0.375m && \text{given} \\ 25n &= 37.5m && \text{multiply both sides by 100} \\ 50n &= 75m && \text{multiply both sides by 2} \\ n &= 3m && \text{divide both sides by 25} \end{aligned}$$

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$$2n = 3m \quad \text{divide both sides by 25}$$
$$12n = 18m \quad \text{multiply both sides by 6}$$
$$\frac{12n}{m} = 18 \quad \text{divide both sides by } m$$

The correct answer is A.

•

N and M are each 3-digit integers. Each of the numbers 1, 2, 3, 6, 7, and 8 is a digit of either N or M. What is the smallest possible positive difference between N and M ?

A. 29

B. 49

C. 58

D. 113

E. 131

Arithmetic Applied Problems

Suppose $N > M$. Let's represent N with its digits abc and M with its digits def, where each of a, b, c, d, e, and f is a different one of the numbers 1, 2, 3, 6, 7, or 8. We want to assign these numbers to the letters to minimize the difference between abc and def. To do this, first we minimize the difference between a and d. That is, $a - d$ must equal 1. Thus, $\{a, d\}$ must be either {2, 1}, {3, 2}, {7, 6}, or {8, 7}. Next, to make the larger number abc as close as possible to the smaller number def, we assign the smallest available digit to b and the largest to e. That is, we set $b = 1$ and $e = 8$. Since each digit can occur only once, this means $\{a, d\}$ cannot contain 8 or 1. That means $\{a, d\}$ cannot be {2, 1} or {8, 7}, so it must be either {3, 2} or {7, 6}. Finally, the last two digits f and c must be whichever two digits have not yet been assigned as values of a, b, d, or e. We again minimize the difference between the larger number abc and the smaller number def by assigning the smaller of these two remaining digits to c and the larger to f, so $f > c$. Given these constraints, we find that either $N = 316$ and $M = 287$ or else $N = 712$ and $M = 683$. In either case, $N - M = 29$. Therefore, 29 is the smallest possible positive difference between N and M.

The correct answer is A.

Technique	Percent of Consumers
Television ads	35%
Coupons	22%
Store displays	18%
Samples	15%

•

The table shows partial results of a survey in which consumers were asked to indicate which one of six promotional techniques most influenced their decision to buy a new food product. Of those consumers who indicated one of the four techniques listed, what fraction indicated either coupons or store displays?

A. $\frac{2}{7}$

B. $\frac{1}{3}$

C. $\frac{2}{5}$

D. $\frac{4}{9}$

E. $\frac{1}{2}$

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Arithmetic Percents; Ratio and Proportion

Let T be the total number of consumers who were asked. Then the table shows 35% + 22% + 18% + 15% = 90% of this total, or $0.9T$ consumers. Also, the number of consumers who indicated either coupons or store displays was $0.22T + 0.18T = 0.4T$. Therefore, the number of consumers who indicated either coupons or store displays divided by the number of consumers shown in the table is $\frac{0.4T}{0.9T} = \frac{4}{9}$.

The correct answer is D.

•

If 65 percent of a certain firm's employees are full-time and if there are 5,100 more full-time employees than part-time employees, how many employees does the firm have?

A. 8,250

B. 10,200

C. 11,050

D. 16,500

E. 17,000

Algebra Simultaneous Equations

Let T represent the total number of employees the firm has; F, the number of full-time employees; and P, the number of part-time employees. Since 65 percent of the firm's employees are full-time, it follows that 35 percent are part-time. Thus, $F = 0.65T$ and $P = 0.35T$. However, $F = P + 5{,}100$, so $0.65T = 0.35T + 5{,}100$, so $0.30T = 5{,}100$, and $T = \frac{5{,}100}{0.30} = 17{,}000$.

The correct answer is E.

•

The cost C, in dollars, to remove p percent of a certain pollutant from a pond is estimated by using the formula $C = \frac{100{,}000p}{100-p}$. According to this estimate, how much more would it cost to remove 90 percent of the pollutant from the pond than it would cost to remove 80 percent of the pollutant?

A. $500,000

B. $100,000

C. $50,000

D. $10,000

E. $5,000

Algebra; Arithmetic Simplifying Algebraic Expressions; Operations on Rational Numbers

Removing 90% of the pollutant from the pond would cost $\frac{(100{,}000)(90)}{100-90} = \frac{9{,}000{,}000}{10} = 900{,}000$ dollars, and removing 80% of the pollutant would cost $\frac{(100{,}000)(80)}{100-80} = \frac{8{,}000{,}000}{20} = 400{,}000$ dollars. The difference is, then, $900,000 − $400,000 = $500,000.

The correct answer is A.

•

If $xy \neq 0$ and $x^2y^2 - xy = 6$, which of the following could be y in terms of x?

I. $\frac{1}{2x}$

II. $-\frac{2}{}$

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II. $-\frac{2}{x}$

III. $\frac{3}{x}$

A. I only

B. II only

C. I and II

D. I and III

E. II and III

Algebra Second-Degree Equations

$$x^2y^2 - xy = 6 \text{ given}$$
$$x^2y^2 - xy - 6 = 0 \text{ subtract 6 from both sides}$$
$$(xy+2)(xy-3) = 0 \text{ factor}$$

So, $xy + 2 = 0$, which means $xy = -2$ and $y = -\frac{2}{x}$, or $xy - 3 = 0$, which means that $xy = 3$ and $y = \frac{3}{x}$. Thus, y in terms of x could be given by the expressions in II or III.

The correct answer is E.

•

If the positive number d is the standard deviation of n, k, and p, then the standard deviation of $n+1$, $k+1$, and $p+1$ is

A. $d+3$

B. $d+1$

C. $6d$

D. $3d$

E. d

Arithmetic Statistics

To find the standard deviation of the numbers in a set, we find their arithmetic mean, then find the differences between the mean and each number in the set, then square each difference, then average the squared differences, and finally find the nonnegative square root of this average. Let x be the mean of n, k, and p. Then $x+1$ is the mean of $n+1$, $k+1$, and $p+1$. The differences between x and each of n, k, and p exactly equal the differences between $x+1$ and each of $n+1$, $k+1$, and $p+1$, respectively. Thus, the squares of the differences, the average of the squared differences, and the nonnegative square root of that average are exactly the same for $\{n, k, p\}$ as for $\{n+1, k+1, p+1\}$. Therefore, the standard deviation d of n, k, and $p$is also the standard deviation of $n+1$, $k+1$, and $p+1$.

The correct answer is E.

•

In a certain high school, 80 percent of the seniors are taking calculus, and 60 percent of the seniors who are taking calculus are also taking physics. If 10 percent of the seniors are taking neither calculus nor physics, what percent of the seniors are taking physics?

A. 40%

B. 42%

C. 48%

D. 58%

E. 80%

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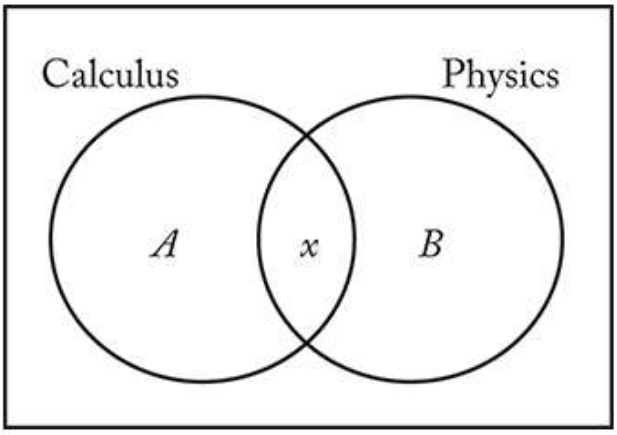
Arithmetic Sets

Determine what percent of the seniors at a certain high school are taking physics.



Consider the Venn diagram on the previous page, where A represents the percent of the seniors who are taking calculus only, B represents the percent of the seniors who are taking physics only, and x is the percent of the seniors who are taking both calculus and physics. As stated, 80 percent of the seniors are taking calculus, so $A + x = 0.8$, and 60 percent of the seniors taking calculus are taking physics, so $x = (0.8)(0.6) = 0.48$. Also, $A + x + B = 0.9$ since 10 percent of the seniors are taking neither calculus nor physics. It follows that $B = 0.9 - (A + x) = 0.9 - 0.8 = 0.1$. Therefore, the percent of the seniors who are taking physics is given by $B + x = 0.1 + 0.48 = 0.58$ or 58 percent.

The correct answer is D.

•

If the units digit of $\frac{5,610.37}{10^k}$ is 6, what is the value of k?

A. 3

B. 2

C. 1

D. −1

E. −2

Arithmetic Exponents

As shown in the table below, the units digit of $\frac{5,610.37}{10^k}$ is 6 when $k = 2$.

$$\frac{5,610.37}{10^{-2}} = \frac{5,610.37}{0.01} = 561,03\mathbf{7}$$

$$\frac{5,610.37}{10^{-1}} = \frac{5,610.37}{0.1} = 56,10\mathbf{3}.7$$

$$\frac{5,610.37}{10^{0}} = \frac{5,610.37}{1} = 5,61\mathbf{0}.37$$

$$\frac{5,610.37}{10^{1}} = \frac{5,610.37}{10} = 56\mathbf{1}.037$$

$$\frac{5,610.37}{10^{2}} = \frac{5,610.37}{100} = 5\mathbf{6}.1037$$

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$$\frac{5{,}610.37}{10^3} = \frac{5{,}610.37}{1{,}000} = 5.61037$$

The correct answer is B.

•

Three printing presses, R, S, and T, working together at their respective constant rates, can do a certain printing job in 4 hours. S and T, working together at their respective constant rates, can do the same job in 5 hours. How many hours would it take R, working alone at its constant rate, to do the same job?

A. 8

B. 10

C. 12

D. 15

E. 20

Algebra Applied Problems

Let r be the portion of the job that printing press R, working alone, completes in 1 hour, and let s and t be the corresponding portions, respectively, for printing press S and printing press T. From the given information, it follows that $r + s + t = \frac{1}{4}$ and $s + t = \frac{1}{5}$. Subtracting these two equations gives $r = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$. It follows that printing press R, working alone, will complete $\frac{1}{20}$ of the job in 1 hour, and therefore printing press R, working alone, will complete the job in 20 hours.

The correct answer is E.

Results of a Poll

Company	Number Who Own Stock in the Company
AT&T	30
IBM	48
GM	54
FORD	75
US Air	83

•

In a poll, 200 subscribers to *Financial Magazine* X indicated which of five specific companies they own stock in. The results are shown in the table above. If 15 of the 200 own stock in both IBM and AT&T, how many of those polled own stock in neither company?

A. 63

B. 93

C. 107

D. 122

E. 137

Arithmetic Sets; Interpretation of Tables

Since 48 of the 200 subscribers polled own stock in IBM and 15 of these 48 subscribers also own stock in AT&T, it follows that 48 − 15 = 33 subscribers own stock in IBM but not in AT&T, as shown in the Venn diagram below. Similarly, 30 − 15 = 15 of the 200 subscribers own stock in AT&T but not in IBM. Therefore, if n is the number of those polled who do not own stock in either IBM or AT&T, then $33 + 15 + 15 + n = 200$, or $n = 137$.

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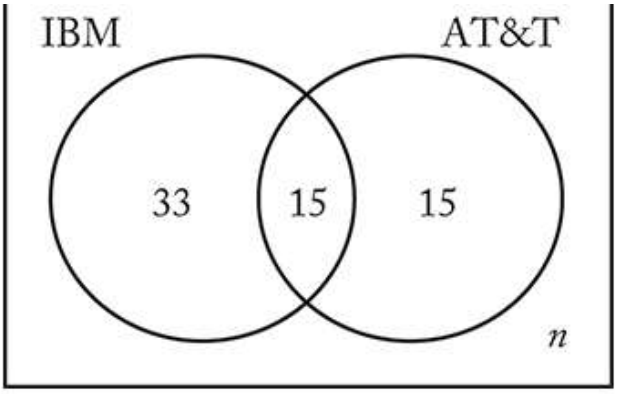


The correct answer is E.

•

The sum of all the integers k such that $-26 < k < 24$ is

A. 0

B. −2

C. −25

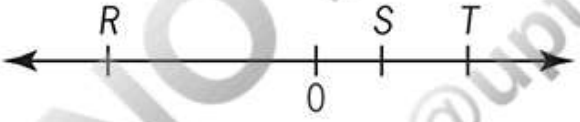
D. −49

E. −51

Arithmetic Operations on Integers

In the sum of all integers k such that $-26 < k < 24$, the positive integers from 1 through 23 can be paired with the negative integers from −1 through −23. The sum of these pairs is 0 because $a + (-a) = 0$ for all integers a. Therefore, the sum of all integers k such that $-26 < k < 24 is -25 + (-24) + (23)(0) = -49$.

The correct answer is D.



•

The number line shown contains three points R, S, and T, whose coordinates have absolute values r, s, and t, respectively. Which of the following equals the average (arithmetic mean) of the coordinates of the points R, S, and T?

A. s

B. $s + t - r$

C. $\frac{r - s - t}{3}$

D. $\frac{r + s + t}{3}$

E. $\frac{s + t + r}{3}$

Arithmetic Absolute Value; Number Line

Because point R is to the left of 0 on the number line, the coordinate of R is negative. It is given that r is the absolute value of the coordinate of R, and so the coordinate of R is $-r$. Because points S and T are to the right of 0 on the number line, their coordinates are positive. It is given that s and t are the absolute values of the coordinates of S and T, and so the coordinates of S and T are s and t. The arithmetic mean of the coordinates of R, S, and T is $\frac{s + t - r}{3}$.

The correct answer is E.

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The correct answer is E.

•

Mark and Ann together were allocated n boxes of cookies to sell for a club project. Mark sold 10 boxes less than n and Ann sold 2 boxes less than n. If Mark and Ann have each sold at least one box of cookies, but together they have sold less than n boxes, what is the value of n?

A. 11

B. 12

C. 13

D. 14

E. 15

Algebra Inequalities

Mark sold $n-10$ boxes and Ann sold $n-2$ boxes. Because each person sold at least one box, it follows that $n-10 \geq 1$ and $n-2 \geq 1$, which implies that $n \geq 11$. On the other hand, together they sold less than n boxes, so $(n-10)+(n-2)<n$, which implies that $n<12$. Therefore, n is an integer such that $n \geq 11$ and $n<12$, which implies that $n=11$.

$$\begin{array}{r} 3P5 \\ +4QR \\ \hline 8S4 \end{array}$$

The correct answer is A.

•

In the correctly worked addition problem shown, P, Q, R, and S are digits. If $Q=2P$, which of the following could be the value of S?

A. 3

B. 4

C. 5

D. 7

E. 9

Arithmetic Place Value

Step 1: Analysis of the units column.

By considering the sum of the unit digits, $5+R=4$ or $5+R=14$. Since $R \geq 0$, we have $5+R \geq 5$, and thus $5+R=4$ is not possible. Therefore, $5+R=14$. It follows that $R=9$ and 1 is carried to the tens column.

Step 2: Analysis of the tens and hundreds columns.

Since 3 + 4 = 7 and the hundreds digit of 8S4 is 8, it follows that 1 must have been carried from the tens column to the hundreds column. Therefore, the sum of the tens digits must be $S+10$, and hence $1+P+Q=S+10$, where the 1 was carried from the units column.

Step 3: Apply the assumption $Q=2P$ to rule out answer choices B, C, and D.

Substituting $2P$ for Q in the equation $1+P+Q=S+10$ gives $1+P+2P=S+10$, which can be rewritten as $S=3P-9=3(P-3)$. This shows that S is divisible by 3, and therefore, answer choices B, C, and D CANNOT be correct.

Step 4: Show that $S=9$ is not possible.

If $S=9$, then we have $9=3(P-3)$. Solving this equation gives $P=6$. However, from $Q=2P$, it follows that $Q=12$, which is not a digit. Therefore, $S=9$ is not possible and answer choice E is NOT correct.

The only possibility remaining is $S=3$. Although it is not necessary to verify that $S=3$ is consistent, one can show that $S=3$ leads to the following for the sum:

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$$\begin{array}{r} 345 \\ +\,489 \\ \hline 834 \end{array}$$

The correct answer is A.

•

A certain high school has 5,000 students. Of these students, x are taking music, y are taking art, and z are taking both music and art. How many students are taking neither music nor art?

A. $5{,}000 - z$

B. $5{,}000 - x - y$

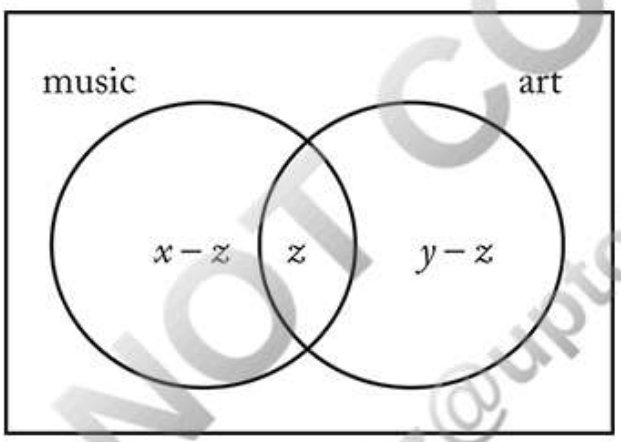
C. $5{,}000 - x + z$

D. $5{,}000 - x - y - z$

E. $5{,}000 - x - y + z$

Algebra Sets

Since x students are taking music, y students are taking art, and z students are taking both music and art, the number of students taking only music is $x - z$, and the number of students taking only art is $y - z$, as illustrated by the Venn diagram below.



Therefore, the number of students taking neither music nor art is
$5{,}000 - [(x - z) + z + (y - z)] = 5{,}000 - x - y + z$.

The correct answer is E.

•

Each person who attended a company meeting was either a stockholder in the company, an employee of the company, or both. If 62 percent of those who attended the meeting were stockholders and 47 percent were employees, what percent were stockholders who were NOT employees?

A. 34%

B. 38%

C. 45%

D. 53%

E. 62%

Arithmetic Sets

Let M represent the number of meeting attendees. Then, since 62% of M or $0.62M$ were stockholders and 47% of M or $0.47M$ were employees, it follows that $0.62M + 0.47M = 1.09M$ were either stockholders, employees, or both. Since $1.09M$ exceeds M, the excess $1.09M - M = 0.09M$ must be the number of attendees who were both stockholders and employees, leaving the rest $0.62M - 0.09M = 0.53M$, or 53%, of the meeting attendees to be stockholders but not employees.

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stockholders but not employees.

The correct answer is D.

Accounts	Amount Budgeted	Amount Spent
Payroll	$110,000	$117,000
Taxes	40,000	42,000
Insurance	2,500	2,340

•

If M is the least common multiple of 90, 196, and 300, which of the following is not a factor of M?

A. 600

B. 700

C. 900

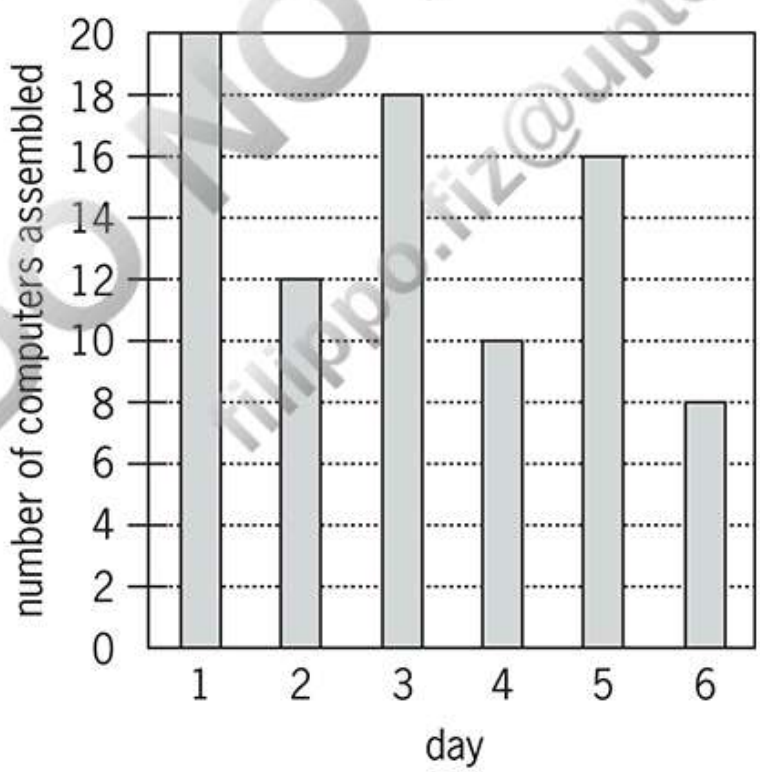
D. 2,100

E. 4,900

Arithmetic Applied Problems

Let x be the least common multiple of 90, 196, and 300. To find x, first use prime factorization to express 90, 196, and 300 as products of prime factors: 90 = 2 × 3 × 3 × 5, while 196 = 2 × 2 × 7 × 7, and 300 = 2 × 2 × 3 × 5 × 5. Then x is the smallest number with a prime factorization including at least as many 2s, 3s, 5s, and 7s as are in the prime factorization of any one of 90, 196, and 300: that is, $x = 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7$. Since this prime factorization includes two 2s, two 3s, two 5s, and two 7s, any factor of x will have a prime factorization that includes only at most two 2s, at most two 3s, at most two 5s, and at most two 7s. Thus, 600 = 2 × 2 × 2 × 3 × 5 × 5 can't be a factor of x, because 600 has three 2s among its prime factors whereas x only has two 2s. But x's factors do include 700 (2 × 2 × 5 × 5 × 7), 900 (2 × 2 × 3 × 3 × 5 × 5), 2,100 (2 × 2 × 3 × 5 × 5 × 7), and 4,900 (2 × 2 × 5 × 5 × 7 × 7), because x includes at least as many 2s, 3s, 5s, and 7s in its prime factorization as any one of these numbers does.

The correct answer is A.



•

The graph shows the number of computers assembled during each of 6 consecutive days. From what day to the next day was the percent change in the number of computers assembled the greatest in magnitude?

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A. From Day 1 to Day 2

B. From Day 2 to Day 3

C. From Day 3 to Day 4

D. From Day 4 to Day 5

E. From Day 5 to Day 6

Arithmetic Percents

The following table shows the percent change from each day to the next and the magnitude of the percent change.

Time Period	Percent Change	Magnitude of Percent Change
From Day 1 to Day 2	$\left(\frac{12-20}{20} \times 100\right)\% = \left(-\frac{8}{20} \times 100\right)\% = -40\%$	40
From Day 2 to Day 3	$\left(\frac{18-12}{12} \times 100\right)\% = \left(\frac{6}{12} \times 100\right)\% = 50\%$	50
From Day 3 to Day 4	$\left(\frac{10-18}{18} \times 100\right)\% = \left(-\frac{8}{18} \times 100\right)\% \approx -40\%$	44
From Day 4 to Day 5	$\left(\frac{16-10}{10} \times 100\right)\% = \left(\frac{6}{10} \times 100\right)\% = 60\%$	60
From Day 5 to Day 6	$\left(\frac{8-16}{16} \times 100\right)\% = \left(-\frac{8}{16} \times 100\right)\% = -50\%$	50

The correct answer is D.

•

If $n = 20! + 17$, then n is divisible by which of the following?

I. 15

II. 17

III. 19

A. None

B. I only

C. II only

D. I and II

E. II and III

Arithmetic Properties of Numbers

Because 20! is the product of all integers from 1 through 20, it follows that 20! is divisible by each integer from 1 through 20. In particular, 20! is divisible by each of the integers 15, 17, and 19. Since 20! and 17 are both divisible by 17, their sum is divisible by 17, and hence the correct answer will include Statement II. If n were divisible by 15, then $n - 20!$ would be divisible by 15. But $n - 20! = 17$, and 17 is not divisible by 15.

Therefore, the correct answer does not include Statement I. If n were divisible by 19, then $n - 20!$ would be divisible by 19. But $n - 20! = 17$, and 17 is not divisible by 19. Therefore, the correct answer does not include Statement III.

The correct answer is C.

•

Exchange Rates in a Particular Year

$1 = 5.3 francs

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$1 = 1.6 marks

An American dealer bought a table in Germany for 480 marks and sold the same table in France for 2,385 francs. What was the dealer's gross profit on the two transactions in dollars?

A. $0

B. $50

C. $100

D. $150

E. $200

Arithmetic Measurement Conversion

Given that $1 = 1.6 marks, it follows that $\frac{\$1}{1.6} = 1$ mark and 480 marks $= 480\left(\frac{\$1}{1.6}\right) = \$300$. Similarly, given that $1 = 5.3 francs, it follows that $\frac{\$1}{5.3} = 1$ franc and 2,385 francs = 2,385 $\left(\frac{\$1}{5.3}\right) = \$450$. The gross profit on the two transactions is $450 − $300 = $150.

The correct answer is D.

•

The product of two negative numbers is 160. If the lesser of the two numbers is 4 less than twice the greater, what is the greater number?

A. –20

B. –16

C. –10

D. – 8

E. – 4

Algebra Second-Degree Equations

Let x and y be the two numbers, where x is the lesser of the two numbers and y is the number desired. From the given information, it follows that $xy = 160$ and $x = 2y - 4$, from which it follows that $(2y - 4)y = 160$. Dividing both sides of the last equation by 2 gives $(y - 2)y = 80$. Thus, 80 is to be written as a product of two negative numbers, one that is 2 less than the other. Trying simple factorizations of 80 quickly leads to the value of y : $(-40)(-2) = 80$, (–20)(–4) = 80, (–10)(–8) = 80. Therefore, $y = -8$. Note that because –8 is one of the answer choices, it is not necessary to ensure there are no other negative solutions to the equation $(y - 2)y = 80$.

Alternatively, $(y - 2)y = 80$ can be written as $y^2 - 2y - 80 = 0$. Factoring the left side gives $(y + 8)(y - 10) = 0$, and $y = -8$ is the only negative solution.

The correct answer is D.

•

According to a certain estimate, the depth $N(t)$, in centimeters, of the water in a certain tank at t hours past 2:00 in the morning is given by $N\left(t\right) = -20(t - 5)^2 + 500$ for $0 \le t \le 10$. According to this estimate, at what time in the morning does the depth of the water in the tank reach its maximum?

A. 5:30

B. 7:00

C. 7:30

D. 8:00

E. 9:00

Algebra Functions

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Algebra Functions

When $t = 5$, the value of $-20(t-5)^2 + 500$ is 500. For all values of t between 0 and 10, inclusive, except $t = 5$, the value of $-20(t-5)^2$ is negative and $-20(t-5)^2 + 500 < 500$. Therefore, the tank reaches its maximum depth 5 hours after 2:00 in the morning, which is 7:00 in the morning.

The correct answer is B.

•

After driving to a riverfront parking lot, Bob plans to run south along the river, turn around, and return to the parking lot, running north along the same path. After running 3.25 miles south, he decides to run for only 50 minutes more. If Bob runs at a constant rate of 8 minutes per mile, how many miles farther south can he run and still be able to return to the parking lot in 50 minutes?

A. 1.5

B. 2.25

C. 3.0

D. 3.25

E. 4.75

Algebra Applied Problems

After running 3.25 miles south, Bob has been running for $(3.25 \text{ miles})\left(8\frac{\text{minutes}}{\text{mile}}\right) = 26$ minutes. Thus, if t is the number of additional minutes that Bob can run south before turning around, then the number of minutes that Bob will run north, after turning around, will be $t + 26$. Since Bob will be running a total of 50 minutes after the initial 26 minutes of running, it follows that $t + (t + 26) = 50$, or $t = 12$. Therefore, Bob can run south an additional $\frac{12 \text{ minutes}}{8\frac{\text{minutes}}{\text{mile}}} = 1.5$ miles before turning around.

The correct answer is A.

•

Alex deposited x dollars into a new account that earned 8 percent annual interest, compounded annually. One year later Alex deposited an additional x dollars into the account. If there were no other transactions and if the account contained w dollars at the end of two years, which of the following expresses x in terms of w?

A. $\frac{w}{1 + 1.08}$

B. $\frac{w}{1.08 + 1.16}$

C.

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